Analysis of Dempsey Center Client Survey Data (2022–2025)

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1. Introduction

This project examines annual client survey data from the Dempsey Center spanning the years 2022 to 2025. The primary objective is to evaluate client satisfaction, the perceived impact of provided services, and the likelihood of clients recommending the Center to others.

In addition, this analysis investigates the relationship between client age, reported barriers to access, and service selection. By reviewing these factors, we aim to understand shifting age trends, the most common challenges faced by clients, and how services are utilized across the population. A key focus was determining whether client age influences the types of services requested.

The research began with a general summary of the data, followed by advanced statistical methods to identify deeper patterns. Our findings indicate that age is not a determining factor in service selection. Furthermore, the data suggests that while clients encounter various barriers, these challenges do not negatively impact their overall satisfaction or continued use of the Center's services. These results confirm the resilience of the Center's service model and provide insight into how best to support clients regardless of demographic change.

2. Data Description and Preprocessing

2.1 Data Sources

The dataset consists of annual client survey responses collected from 2022 to 2025. Each survey captures client satisfaction, perceived impact of services, service usage, and demographic characteristics.

2.2 Data Cleaning

Preprocessing involved removing duplicate header rows from raw files, standardizing column formats across years, and resolving missing or inconsistent values. Records from all four years were merged into a unified dataset with a year variable added to support temporal analysis.

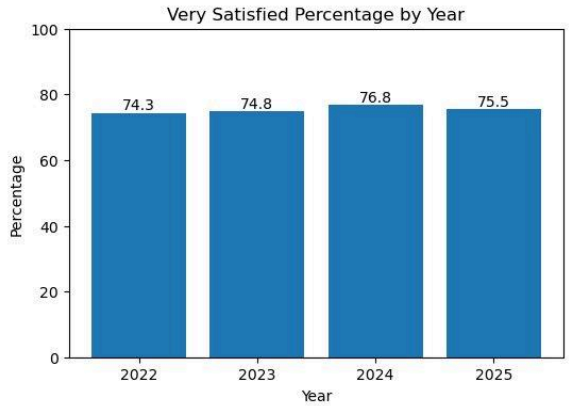
For the deeper analysis, we narrowed focus to three primary feature sets: age, services, and barriers. Service selections were encoded as binary vectors, arrays of 0s and 1s, enabling mathematical analysis of service usage patterns across age groups.

2.3 Data Overview

After cleaning, the dataset contains 269 responses from 2022, 298 from 2023, 293 from 2024, and 319 from 2025.

3. Basic Summary of Data

3.1 Overview of Key Metrics



Overall satisfaction is consistently high across all four years, with approximately 74–77% of respondents reporting being

"Very Satisfied" and over 85–90% reporting being either satisfied or very satisfied.

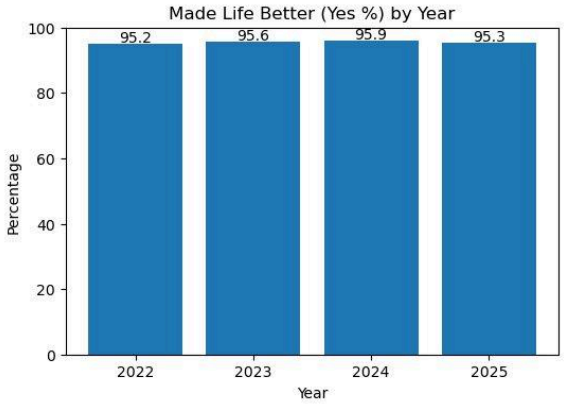
3.2 Distribution of Satisfaction

Satisfaction responses are heavily skewed toward positive outcomes. The majority of respondents reported being Very Satisfied, with very few negative responses, indicating that high satisfaction is not just an average trend but a consistent pattern across the population.

3.3 Overall Satisfaction Trends

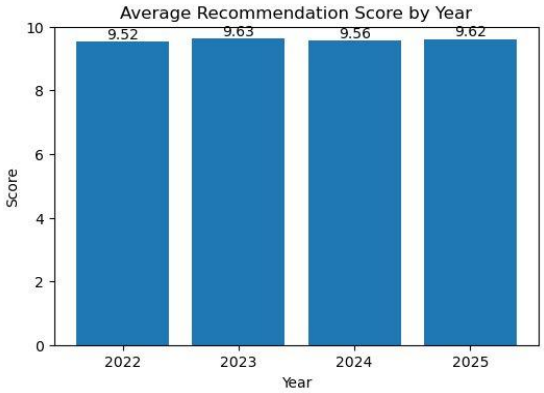
The percentage of respondents reporting Very Satisfied remains stable across all four years, ranging from approximately 74% to 77%. This consistency suggests the Dempsey Center has maintained a reliable level of service quality over time.

3.4 Perceived Impact of Services



Approximately 95% of respondents reported that services made their life better, with minimal variation across years.

3.5 Likelihood to Recommend



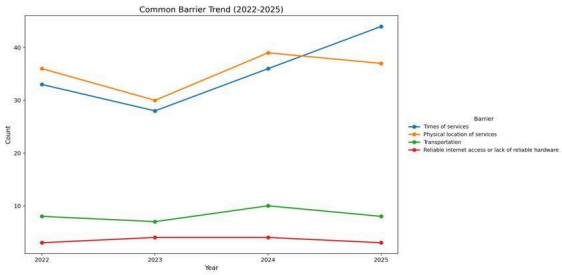
The average recommendation score ranges from 9.5 to 9.6 out of 10, reflecting that clients are highly willing to recommend the Center's services to others.

3.6 Key Takeaways from Summary

These metrics paint a clear picture: the Dempsey Center consistently delivers highly valued services. Satisfaction is high, perceived impact is near-universal, and clients are strong advocates for the Center. However, these aggregate statistics mask potential differences between client subgroups, a gap that our later in-depth analysis addresses directly.

4. In-Depth Analysis

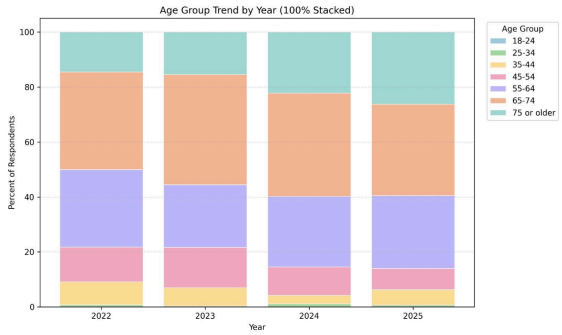
4.1 Barriers



Service timing and physical location are the two most frequently reported barriers, and both have grown more common over the four-year period. After a dip in 2023, both barriers rose, with service timing emerging

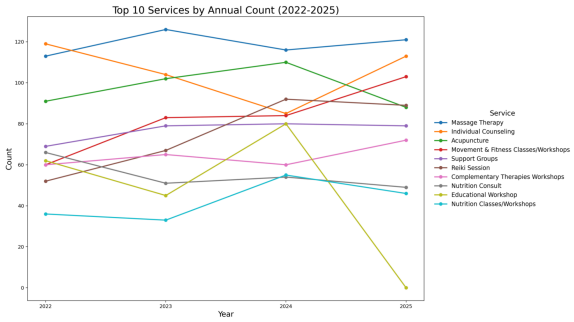
as the top concern by 2025. Other barriers such as transportation and lack of internet or computer access remained low throughout. These findings point to scheduling and geography as the primary friction points for clients to access the Center's services.

4.2 Age



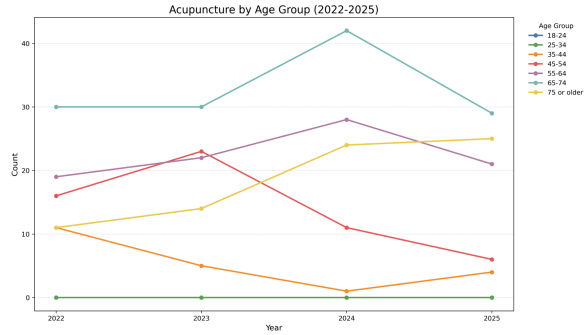
The Dempsey Center's client base is aging. The proportion of clients in the 65–74 and 75 or older age groups has grown across all four years, while younger age groups have remained small. By 2025, the majority of respondents fall into these older categories.

4.3 Services Overview

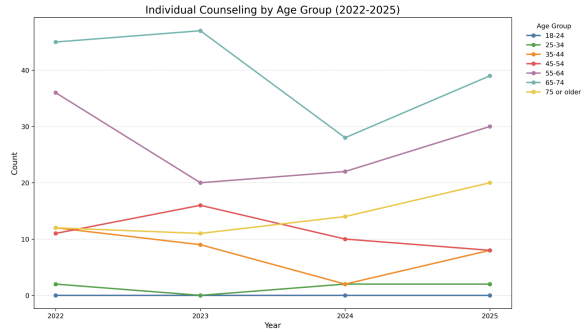


The Top 10 Services by Annual Count graph shows that massage therapy and individual counseling are the most popular programs, consistently reaching over 100 visits per year. Other core services like acupuncture and movement or fitness classes also maintain strong participation, indicating a balanced client interest in both physical and emotional support.

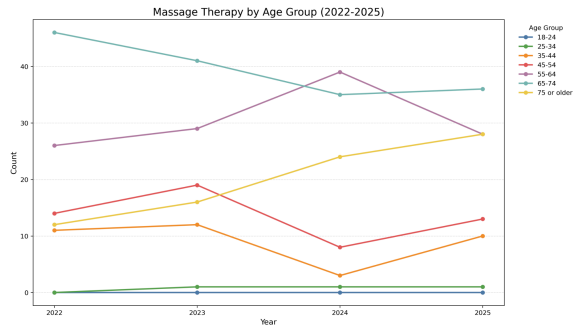
4.4 Popular services by Age Group



Acupuncture is used almost exclusively by clients over 55, with the 65–74 group consistently leading. The 75 or older group was the only one showing steady growth, rising from 11 to 25 over the study period, while the 45–54 group declined.

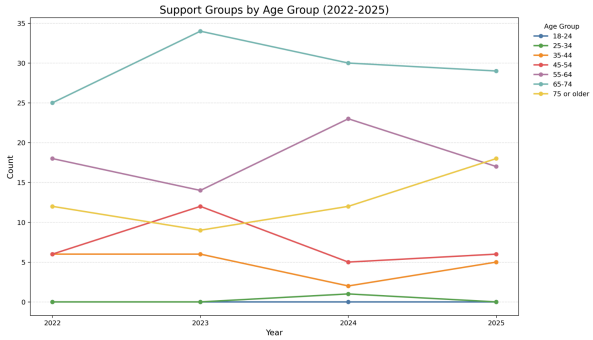


Individual counseling follows a similar age concentration, with the 65–74 group leading but fluctuating across years. The 75 or older group showed consistent growth, while clients under 45 are low.



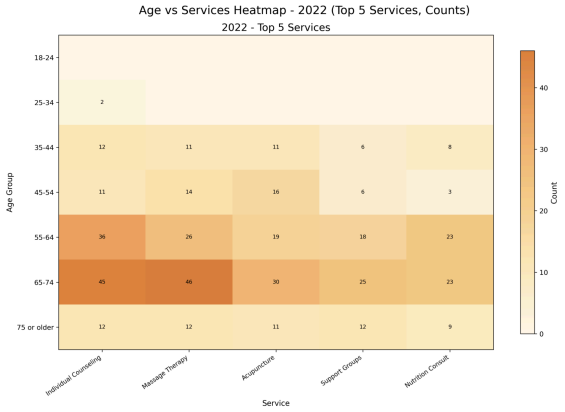
Massage therapy shows a shift within the older age. The 65–74 group declined from over the period while the 75 or older group grew, nearly closing the gap by 2025.

Movement and fitness classes show the strongest overall growth of any service. The 65–74 group grew steadily, and the 75 or older group more than doubled. Clients under 45 were largely absent throughout.

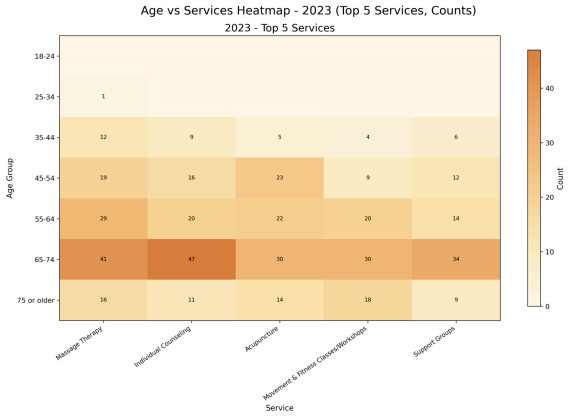


Support group attendance is dominated by the 65–74 group, though their numbers declined slightly from a peak of 34 in 2023 to 29 in 2025. The 75 or older group grew steadily, consistent with the broader trend of the oldest clients becoming an increasingly larger share of usage across all services.

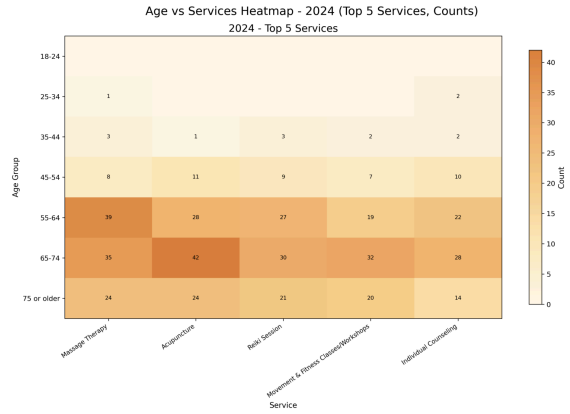
4.5 Services and Age Correlation



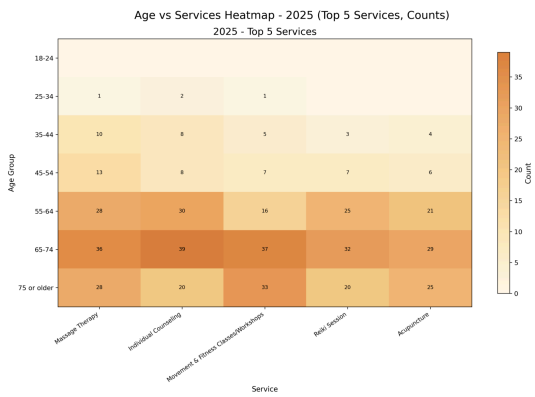
The 2022 heatmap shows usage heavily concentrated in the 65–74 group, who led across all five services with individual counseling and massage therapy as their top choices at 45 and 46 sessions respectively. The 55–64 group was a distant second, and clients under 45 contributed almost nothing.



The 2023 pattern is largely similar, with the 65–74 group again dominating. The most notable change is a stronger showing from the 45–54 group in acupuncture, suggesting brief growing interest among middle-aged clients.



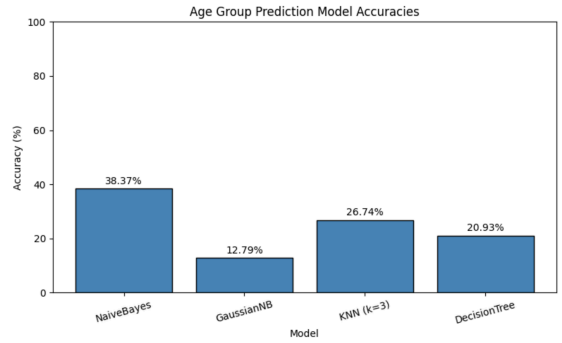
By 2024 the service mix shifted, with acupuncture and reiki replacing nutrition consult and individual counseling in the top five. The 55–64 group lead in massage therapy at 39 sessions, overtaking the 65–74 group for the first time.



The 2025 heatmap shows the broadest distribution across older age groups of any year, with the 75 or older group recording its strongest showing yet in movement and fitness classes.

Given the relatively small sample size, these patterns should be interpreted as directional rather than definitive, and will become more reliable as the cumulative dataset grows.

4.6 Services Pattern



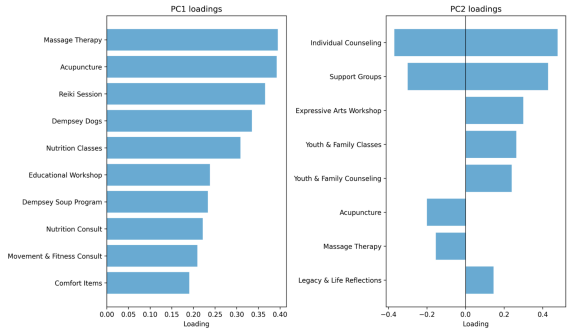
To explore whether a client's service usage pattern could predict their age group, each client was represented as a binary vector of services used, and four models were tested: Naive Bayes, Gaussian Naive Bayes, KNN (k=3), and Decision Tree.

Naive Bayes performed best at 38.37%, which is expected since it handles binary feature data well. However, even this result is only modestly above random chance for a seven-class problem.

Gaussian Naive Bayes scored the lowest at 12.79%, near or below random chance, likely because it assumes continuous normally distributed features rather than binary data.

KNN (k=3) achieved 26.74% and Decision Tree 20.93%, both falling in the middle of the range. Neither model found consistent patterns strong enough to reliably separate age groups.

Taken together, the low accuracy across all four models confirms that service usage patterns are not a reliable predictor of age group, suggesting that other segmentation approaches may be more informative.

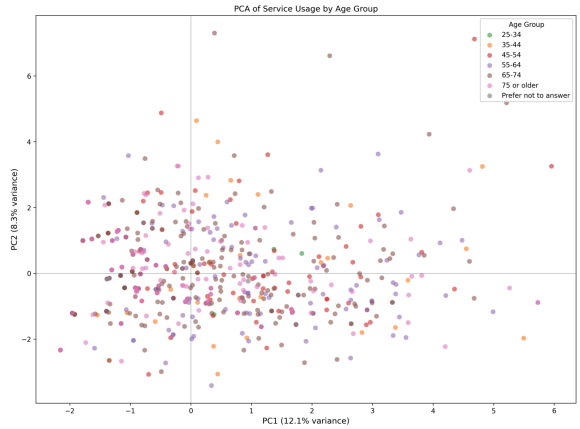


To better understand why the models struggled, PCA was applied to visualize the underlying structure of the service usage data. PCA reduces the high-dimensional binary service vectors into a smaller set of components, each capturing a different dimension of variation across clients.

PC1, which explains 12.1% of the variance, primarily reflects physical engagement volume. The services with the highest loadings are Massage Therapy, Acupuncture, and Reiki Session, suggesting that clients who score high on PC1 tend to be heavy users of physical and body-based therapies.

PC2, explaining 8.3% of the variance, captures a more specific contrast between client types. Individual Counseling and Support Groups load positively while Massage Therapy and Acupuncture load negatively, meaning PC2 separates clients who prefer emotional support services from those who prefer physical therapies.

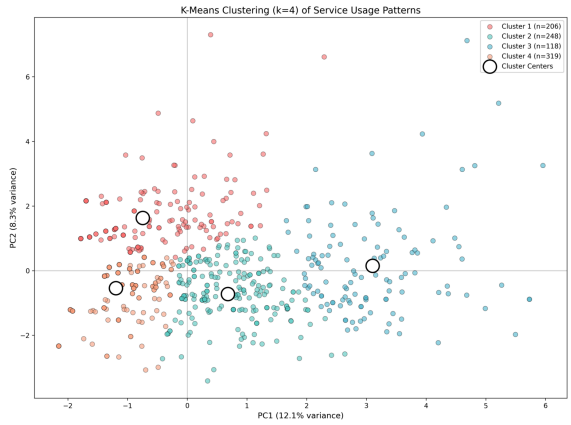
Together the two components account for only about 20% of total variance, which is low but expected given how sparse the binary service data is. This confirms that service usage patterns are distributed across many dimensions with no single dominant structure, helping explain why no classification model was able to predict age group with meaningful accuracy.



The PCA scatter plot projects each client onto the PC1 and PC2 axes, with points colored by age group. The immediate observation is that age groups do not form distinct clusters. Clients of different ages are scattered throughout the same regions with substantial overlap, visually confirming what the classification models already suggested.

The horizontal spread along PC1 shows that some clients use many services while others use very few, regardless of age. The vertical spread along PC2 shows a split between those who prefer emotional support services and those who prefer physical therapies, also independent of age. Neither axis reveals any meaningful separation between age groups.

The scatter plot ultimately tells the same story as the model predictions. The absence of age-based clustering suggests that service choice is driven by something other than age, likely individual health needs or personal preferences, pointing toward the need for a different segmentation approach.



K-Means clustering with k=4 was applied to the PCA reduced data to identify natural groupings in service usage patterns without relying on age as a label. Four clusters emerged with sizes of 206, 248, 118, and 319 clients respectively, suggesting meaningful variation in how clients engage with the center's services.

The cluster plot shows that the four groups occupy partially overlapping but distinguishable regions of the PCA space. Clusters 1 and 4 tend to sit on the left side of the plot, corresponding to lower PC1 scores and suggesting lighter overall service usage. Clusters 2 and 3 spread further to the right, indicating heavier use of multiple services.

The top services by cluster chart gives each group a clearer identity. Cluster 1 is defined by high rates of Individual Counseling and Support Groups at 65% and 59% respectively, marking it as the emotional support group.

Cluster 2 is led by Massage Therapy at 71% and Acupuncture at 63%, identifying it as the physical therapy group.

Cluster 3 shows the highest overall engagement across nearly all services, with Massage Therapy at 87% and Acupuncture at 84%, representing the most active multi-service users.

Cluster 4 has uniformly low usage rates across all services, with no single service exceeding 30%, suggesting a group of light or infrequent users.

These four clusters succeed where age-based classification failed, revealing that the client naturally organizes into four personas: emotional support seekers, physical therapy users, highly engaged multi-service users, and light users. This behavioral segmentation is more meaningful than age as a framework to understand client needs, and provides a foundation for tailoring services and outreach for groups in future.

5. Results

Across all four years, satisfaction remained consistently high at 74–77% very satisfied, perceived impact held steady at 95%, and the average recommendation score stayed near 9.6 out of 10, confirming that the center delivers reliable, high-quality services to its clients.

The client population is aging. The 65–74 and 75 or older age groups have grown steadily. Scheduling and physical location emerged as the two most frequently reported barriers, both trending upward over the study period.

The core question of whether service usage patterns could predict age group was answered clearly in the negative. All four classification models performed poorly, with the best result at 38.37%. PCA confirmed that age groups overlap heavily in service usage space, and K-Means clustering revealed a more meaningful segmentation of four behaviorally distinct client groups: emotional support seekers, physical therapy users, highly engaged multi-service users, and light users.

6. Limitations

The data is entirely self-reported and may be subject to recall bias. Survey wording and service naming also varied slightly across years, requiring normalization that may have introduced minor inconsistencies in cross-year comparisons.

The sample size is relatively small for the depth of analysis performed. With annual response counts between 269 and 319, many subgroup cells contain single-digit counts, and the absence of client identifiers limits all analysis to cross-sectional snapshots rather than longitudinal tracking.

Finally, PCA retains only about 20% of total variance in the first two components, and the k=4 clustering solution was chosen based on interpretability rather than a strict optimality criterion, introducing some subjectivity into the segmentation results.

7. Recommendations

7.1 Expand Access

Scheduling and physical location were the two most frequently reported barriers, both trending upward over the study period. Expanding service hours would directly address the time barrier. Given the steady growth of the 65 and older population, the center should consider targeted outreach designed specifically for this demographic.

7.2 Optimize Services

Resources should be concentrated on consistently high-demand services such as massage therapy, individual counseling, and acupuncture, which maintained strong participation across all years. Low-volume services that show declining or negligible usage should be reviewed for redesign or phase-out to allow more efficient allocation of staff and facilities.

7.3 Improve Feedback Loop

Current satisfaction scores are high, which while positive, limits the ability to identify specific areas for improvement. Future surveys should include more targeted questions around service quality, barrier severity, and unmet needs. Actively soliciting critical feedback, particularly from clients who report barriers or lower satisfaction, would provide more actionable insight for continuous improvement.

8. Future Work

8.1 Qualitative Analysis

The current analysis focused primarily on structured survey responses, leaving open-ended qualitative data unexplored. Future work should apply sentiment analysis and topic modeling to text responses, with particular attention to negative feedback, which represents the most actionable signal for identifying gaps in service delivery.

8.2 Improved Segmentation

The four behavioral clusters identified here provide a starting point but could be refined with richer data. Incorporating variables such as cancer diagnosis type, length of engagement, and service frequency would allow for more clinically meaningful personas, and a growing cumulative dataset will eventually enable longitudinal tracking that the current cross-sectional data cannot support.